

Review

LRRK2 and idiopathic Parkinson's disease

Emily M. Rocha,^{1,4} Matthew T. Keeney,^{1,2,4} Roberto Di Maio,¹ Briana R. De Miranda,³ and J. Timothy Greenamyre^{1,2,*,@}

The etiology of idiopathic Parkinson's disease (iPD) is multifactorial, and both genetics and environmental exposures are risk factors. While mutations in leucine-rich repeat kinase-2 (*LRRK2*) that are associated with increased kinase activity are the most common cause of autosomal dominant PD, the role of *LRRK2* in iPD, independent of mutations, remains uncertain. In this review, we discuss how the architecture of *LRRK2* influences kinase activation and how enhanced *LRRK2* substrate phosphorylation might contribute to pathogenesis. We describe how oxidative stress and endolysosomal dysfunction, both of which occur in iPD, can activate non-mutated *LRRK2* to a similar degree as pathogenic mutations. Similarly, environmental toxicants that are linked epidemiologically to iPD risk can also activate *LRRK2*. In aggregate, current evidence suggests an important role for *LRRK2* in iPD.

LRRK2: a potential link between genetic and idiopathic forms of Parkinson's disease

PD is a complex, multifactorial, progressive disease characterized by motor and non-motor neurological symptoms. The motor symptoms are driven, in part, by the loss of dopaminergic neurons in the substantia nigra (SN) and, in most cases, are accompanied by proteinaceous inclusions, referred to as Lewy bodies and neurites. The age of diagnosis, clinical manifestation, disease progression, and neuropathology vary widely between individuals. Approximately 10% of PD cases are attributed to inherited genetic mutations, while the rest are considered idiopathic or sporadic. Mutations in risk genes elevate the chances of developing PD; however, the phenotypic penetrance (see Glossary) of risk genes is incomplete. The likelihood of developing iPD depends on several factors, including age, the presence of risk alleles, and environmental factors (e.g., **tox-icants** and lifestyle) [1–4]. The G2019S mutation in the *LRRK2* protein is the most common pathogenic mutation, accounting for 1–6% of sporadic and 3–19% of familial PD cases [5]. Additionally, common variants in the *LRRK2* locus are known risk factors for iPD. Indeed, it has been argued that *LRRK2* provides a link between genetic and idiopathic forms of PD [6].

Although *LRRK2* is a low-abundance protein in neurons, its kinase activity is significantly increased in SN dopamine (DA) neurons and microglia of patients with iPD [7]. Sustained elevation in *LRRK2* kinase activity likely contributes to the pathogenesis of iPD: using *LRRK2* kinase inhibitors or *LRRK2* silencing, several investigators have shown prevention of brain pathologies in rodent models of PD [8–11]. While there is a growing consensus that *LRRK2* kinase activity is increased in iPD, how it gets activated and its roles in disease pathogenesis remain unclear.

Clinically, carriers of *LRRK2* mutations who develop PD are largely indistinguishable from patients with late-onset iPD. Despite the clinical similarity, the neuropathology is heterogeneous in carriers of *LRRK2* mutations, even among family members with the same mutation [12]. It is clear that carriers of *LRRK2* mutations exhibit profound nigrostriatal and locus coeruleus degeneration. However, the presence of Lewy pathology is variable: while most carriers of the G2019S mutation exhibit Lewy pathology, a minority of these carriers do not [13]. This variability, even among

Highlights

Mutations in *LRRK2* that are associated with aberrantly enhanced kinase activity are the most common cause of genetic Parkinson's disease (PD). Emerging evidence now points to a role of elevated *LRRK2* kinase activity without *LRRK2* mutations in nongenetic (idiopathic) forms of the disease.

The architecture of *LRRK2* influences kinase activation, and enhanced *LRRK2* substrate phosphorylation might contribute to PD pathogenesis.

Evidence indicates that *LRRK2* kinase activity might be enhanced in idiopathic (i) PD by oxidative stress and/or endolysosomal stress.

Exposure to environmental toxicants linked to iPD by epidemiological studies is also reported to activate *LRRK2* kinase activity.

Two therapeutic approaches targeting *LRRK2*, an antisense oligonucleotide and a kinase inhibitor, are in clinical trials for both carriers of *LRRK2* mutations and those with iPD.

¹Pittsburgh Institute for Neurodegenerative Diseases and Department of Neurology, University of Pittsburgh School of Medicine, Pittsburgh, PA, USA

²Molecular Pharmacology Graduate Program, University of Pittsburgh, Pittsburgh, PA, USA

³Center for Neurodegeneration and Experimental Therapeutics, Department of Neurology, The University of Alabama at Birmingham, Birmingham, AL, USA

⁴These authors contributed equally

*Correspondence: jgreena@pitt.edu (J.T. Greenamyre).
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those with the same mutation, suggests that α -synuclein pathology is not necessary for the typical clinical manifestations of the disease, and disease progression might occur through divergent mechanisms. In this context, it appears likely that the manifestations and progression of PD are influenced by gene-environment interactions, in which environmental factors have a role in both the development of clinical symptoms and the type of neuropathology observed at autopsy. In this review, we discuss the role of LRRK2 in iPD, the mechanisms and toxicants that lead to LRRK2 activation, the cellular consequences of prolonged LRRK2 kinase activity, and potential ways to reduce it therapeutically.

LRRK2: conformational dynamics and kinase activation

LRRK2 is a 2527-amino acid protein with two enzymatic domains: (i) a Ras-of-complex (ROC), C-terminal of the Roc domain (COR) (ROC-COR) GTPase domain; and (ii) a serine/threonine kinase domain (Figure 1). The enzymatic domains are flanked by an N-terminal armadillo domain, ankyrin domain, leucine-rich repeat domain, and a C-terminal WD40 domain (Figure 1). The most frequent pathogenic mutations reside in the ROC-COR GTPase (R1441G/C/H, Y1669C) and kinase domains (G2019S, I2020T) [14], indicating important roles for both enzymatic domains in the pathogenicity of LRRK2-driven PD. Each of the pathogenic mutations results in a toxic gain of function: elevated kinase activity [15]. These facts, together with evidence of elevated LRRK2 kinase activity in iPD [7,16], suggest that elucidating the mechanisms that drive LRRK2 kinase activation will be crucial for understanding the pathogenesis of PD.

Embedded in all kinases is a DFG ψ motif, which controls its catalytic activity [17]. LRRK2 is unique in that it instead has a DYGI sequence (residues 2017–2020), the functional consequence of which being that a hydroxyl moiety on tyrosine²⁰¹⁸ forms a hydrogen-bonding node, keeping LRRK2 locked in its inactive conformation [18]. There are two common pathogenic LRRK2 mutations that lie within this DYGI sequence, G2019S and I2020T, which both result in enhanced kinase activity, because they unleash the inactive conformation of the protein [18]. A recent study demonstrated that the G2019S mutation results in a more flexible kinase domain, which

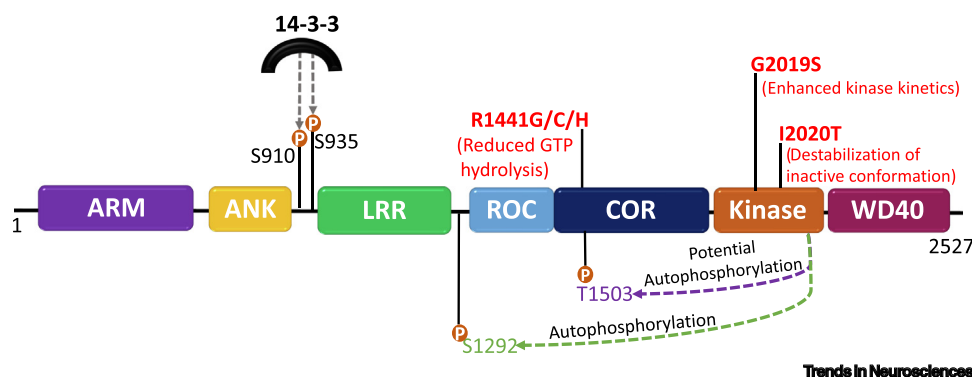


Figure 1. Architecture of the leucine-rich repeat kinase-2 (LRRK2) protein and effects of Parkinson's disease-associated mutations. LRRK2 comprises 2527 amino acids and has seven distinct domains: ARM, armadillo; ANK, ankyrin; LRR, leucine-rich repeat; ROC, Ras of complex; COR C-terminal of ROC; Kinase; and WD40, WD40 domain. Its enzymatic domains are a ROC-COR GTPase domain and a serine/threonine kinase domain. The most common pathogenic mutations (in red) arise in the ROC-COR GTPase or kinase domain of the protein. The proposed mechanism of how each mutation affects the dynamics of LRRK2 is described below each mutation. Also depicted are the autophosphorylation sites, pSer1292 (green) and pThr1503 (purple). The autophosphorylation site pSer1292 is the most widely studied and is commonly used as a readout of kinase activity. Although the Thr1503 autophosphorylation site has been used in several studies as an indicator of kinase activity, the site has yet to be validated in an endogenous system. Phosphorylation sites in the N terminus (pSer910 and pSer935) mediate 14-3-3 binding to LRRK2.

Glossary

Anti-sense oligonucleotide (ASO):

synthetic, short single-stranded oligodeoxynucleotides that are complementary to an mRNA target and thereby can lead to transcript knockdown and, consequently, a reduction in protein expression.

Autophagy: lysosome-dependent degradation system that allows for recycling of macromolecules. It is responsible for removal of unnecessary and/or dysfunctional components of the cell.

Chloroquine: a lysosomotropic drug that becomes protonated in the acidic environment of the lysosome and causes an influx of water into the lysosome, a process referred to as the osmotic effect.

L-leucyl-L-leucine methyl ester

(LLOME): a lysosomotropic agent that results in lysosomal membrane permeabilization and rupture. After endocytosis, it reaches the lysosome, where it interacts with Cathepsin C and causes lysosomal membrane rupture.

Monensin: an antibiotic derived from *Streptomyces cinnamonensis*. It is a monovalent cation ionophore that causes disruption in vesicular transport from the *trans*-Golgi network.

Oxidative stress: pathophysiological event that occurs when the redox state of the cell shifts toward being more oxidized. This shift is the consequence of the accumulation of free radicals and/or loss of antioxidant defense systems.

Penetrance: in the context of human genetics, penetrance refers to the proportion of individuals carrying a particular genetic variant (allele) who also express the phenotype associated with that variant. LRRK2 mutations have incomplete penetrance, meaning that not all individuals carrying a mutation develop PD.

Quinone: class of cyclic organic compounds that contains two carbonyl groups in a six-member unsaturated ring. They are oxidized derivatives of aromatic compounds and are electrophilic molecules that can readily form covalent bonds with nucleophilic moieties, such as sulfhydryl groups of cysteine residues.

Rab small GTPase: members of the Ras superfamily of small G proteins. Approximately 70 Rab proteins have been identified to date and are involved in membrane trafficking, including

enhances its dynamics and likely contributes to elevated kinase activity [19]. This idea was supported by an independent group that reported the structure of full-length inactive human LRRK2 via cryo-electron microscopy [20]. The authors showed that the structure of monomeric inactive wild-type LRRK2 and G2019S LRRK2 are almost identical [20], suggesting that the G2019S mutation does not result in elevated activity by a structural change in LRRK2, but rather through altered kinase kinetics [20]. Thus, the dynamics of the kinase domain might drive LRRK2 activation [19].

Emerging evidence suggests that GTP binding and hydrolysis affects LRRK2 kinase activity, and that G2019S mutation-driven neurodegeneration relies on both GTPase and kinase activity [21]. This points to an intricate relationship between the two enzymatic domains of the LRRK2 protein. Mutations that arise in the ROC-COR GTPase domain affect GTP hydrolysis and result in a prolonged active state [22]. These mutations do not increase the catalytic activity *in vitro*, but do result in increased substrate phosphorylation [15] and increased autophosphorylation at serine¹²⁹² (Ser1292) *in vivo* [23]. Both the ROC-COR GTPase and kinase domains interact with each other, suggesting that GDP and GTP binding influences the conformation or activity of LRRK2 [24]. A 3.5-Å structure of full-length inactive LRRK2 revealed interdomain interactions between the ROC and COR domains [20]. Notably, this interface between the Roc-Cor GTPase domain is the site of many of the LRRK2 mutations that increase the risk for PD [20], suggesting that the interface of the ROC-COR GTPase domain is central to LRRK2 regulation.

Those mutations arising in different enzymatic domains result in elevated kinase activity through unique mechanisms, which highlights the complexity of LRRK2 kinase activation. Given that LRRK2 is a highly dynamic protein, elucidating the crucial regulatory interactions that stabilize LRRK2 in different conformations will lead to a deeper understanding of kinase activation and, possibly, more effective LRRK2 therapeutics to treat PD.

LRRK2: substrates and kinase regulation

The functions of LRRK2 and its paralog LRRK1 are not fully understood. However, it is clear that a subset of **Rab small GTPases** (e.g., Rab8A and Rab10) are physiological substrates of LRRK2 [25] and are distinct from those that are phosphorylated by LRRK1 [26]. Only a subset of LRRK2 substrates, including Rab GTPases, have been validated in endogenous expression systems [25,27]. Given that Rab GTPases regulate vesicular trafficking, LRRK2 appears to function as a master regulator of vesicular trafficking. Upon LRRK2-dependent phosphorylation of a Rab, the interaction of Rab with its GDP dissociation inhibitor (GDI) is prevented, leading to accumulation of the Rab protein in membranes [28]. This heightened accumulation can lead to inefficient Rab cycling, which can result in deficits in both the endolysosomal system and vesicular trafficking. For example, overexpression of G2019S LRRK2 can disrupt endolysosomal trafficking by impairing Rab8A and Rab10, resulting in accumulation of Rab4-positive recycling endosomes and lysosomal dysfunction [29,30]. To help maintain and facilitate efficient Rab cycling, the phosphatase, PPM1H, counteracts LRRK2-mediated Rab phosphorylation by dephosphorylating Rab8A and Rab10 [31]. Elevated phosphorylated Rab10 has been observed in postmortem human brain tissue [7], peripheral immune cells [32], and blood from patients with iPD. Additionally, preliminary results indicated that elevated phosphorylated Rab10 is present in peripheral blood mononuclear cells (PBMCs) in two cohorts of subjects with iPD [33]. These results suggest there is increased LRRK2 kinase activity or decreased phosphatase activity in iPD. However, the status of PPM1H (expression, localization, and activity) in iPD remains to be investigated.

Most of the LRRK2 protein is believed to reside in the cytosol in an inactive state [34] bound to members of the ubiquitously expressed 14-3-3 family of proteins. LRRK2 binding to 14-3-3

vesicle formation and movement along tubules. A subset of Rab GTPases are LRRK2 substrates.

Reduction/oxidation (redox) state: cellular or subcellular state of balance between oxidants and antioxidants in a cell.

Toxicant: substance of natural or synthetic origin that produces transient or permanent damage (toxicity) to cells, tissues, and organs.

proteins at phosphorylation sites in the N-terminal domain (pSer910 and pSer935) of LRRK2 decrease its kinase activity. These phosphorylation sites are currently considered biomarkers for on-target Type I LRRK2 kinase inhibitors, because these inhibitors result in dephosphorylation of Ser910 and Ser935 [35]. This has been validated by several independent groups using kinase inhibitors [21,36–39]. Additionally, using overexpression systems and purified LRRK2 phosphopeptides, more recent data have suggested that LRRK2 binding to 14-3-3 proteins also occurs in the ROC-COR GTPase domain at pSer1444 [40,41]. However, it will be important to validate this phosphorylation site using endogenous expression systems. Loss of 14-3-3: LRRK2 interaction leads to enhanced kinase activity by destabilization of the inactive conformation [41], and it has been reported that some PD-related pathogenic LRRK2 mutations (e.g., R1441G/C/H or Y1699C) that result in increased kinase activity, have less LRRK2:14-3-3 binding [42]. It is unclear whether the G2019S mutation also results in a reduction in LRRK2:14-3-3 binding, because conflicting results have been reported [7,42], possibly because of the use of overexpression versus endogenous expression systems. Nevertheless, the combination of the loss of this regulatory interaction and altered dynamics of the enzymatic domains of LRRK2 could help explain why these mutations result in enhanced kinase activity. In addition, overexpression of 14-3-3 protein can prevent G2019S LRRK2 kinase activity-driven toxicity [43]. Together, these data indicate that 14-3-3 proteins have a major role in LRRK2 kinase regulation and activation. In the context of iPD, phosphorylation of Ser232 on 14-3-3, which leads to its dysfunction, was reported in neuroblastoma cells treated with rotenone or overexpressing α -synuclein [44]. This was also observed in iPD brain tissue [44]. Dissociation of 14-3-3 from LRRK2 has been observed in iPD postmortem human brain tissue [7], indicating that impaired 14-3-3:LRRK2 binding might contribute to the elevated LRRK2 activity observed in iPD.

LRRK2 not only resides in the cytoplasm, but also localizes to many membrane-bound organelles, including endosomes, lysosomes, and mitochondria, as well as microtubules [45–50]. Understanding the subcellular localization of LRRK2 and its interacting proteins will help define its function and how it contributes to PD pathogenesis. At the *trans*-Golgi network (TGN), overexpression of Rab29 recruits and activates LRRK2 by promoting LRRK2 autophosphorylation at Ser1292, which then increases phosphorylation of Rab10 [49–51]. Interestingly, the PARK16 locus, which contains the gene encoding Rab29, has been identified as a risk site for iPD [52]. As such, Rab29 recruitment and activation of LRRK2 at the TGN might be a mechanism that contributes to PD pathogenesis. However, as an important caveat, a recent study reported that endogenous (as opposed to overexpressed) Rab29 does not recruit LRRK2 or stimulate LRRK2 kinase activity [53]. Therefore, Rab29-mediated LRRK2 recruitment and activation, if they occur, might depend on cell type and/or subcellular microdomains with elevated endogenous Rab29 levels. That being said, under some conditions, Rab29 can recruit LRRK2 to stressed lysosomes and stabilizes Rab8 and Rab10 at the lysosomal membrane, preventing lysosomal enlargement [47]. This LRRK2-Rab machinery can also upregulate lysosomal secretion through the Rab effectors, EH Domain-Binding Protein 1 (EHBP1) and EHBP1-like 1 (EHBP1L1) [47,54]. This LRRK2-dependent recruitment of EHBP1 and EHBP1L1 to the lysosome promotes vesicle formation and budding from lysosomal membranes to maintain lysosomal homeostasis [55,56]. It has also been reported that LRRK2 can recruit c-Jun-N-Terminal Kinase (JNK)-Interacting Protein-4 (JIP4) to stressed lysosomes via Rab35 and Rab10, which causes budding of vesicles from stressed lysosomes [46]. Nevertheless, it is clear that LRRK2 is a regulator of endolysosomal function.

LRRK2: endolysosomal dysfunction

The genetic architecture of PD strongly implicates impairments of vesicular trafficking, endolysosomal function, and **autophagy** in PD pathogenesis [6,57]. Aberrant LRRK2 kinase activity is associated with pharmacologically stressed lysosomes. It is possible that genetic mutations that result in

lysosomal dysfunction might also result in increased LRRK2 kinase activity. In addition, several LRRK2-interacting proteins (e.g., auxilin, ATP13A2, LAMP2a, and ArfGAP1) are associated with lysosomal dysfunction [58–62]. Moreover, the D290N VSP35 mutation and certain GBA1 mutations (i.e., N370S and E326K), which impair endolysosomal function, result in increased LRRK2 kinase activity [63,64]. Together, these data suggest that aberrant LRRK2 activity is associated with endolysosomal dysfunction.

Early endosomes are the first cargo vesicle in the endolysosomal pathway (Box 1), and any disruption in maturation or trafficking of early endosomes to late endosomes will likely affect overall lysosomal function. In sporadic Alzheimer's disease, swollen Rab5-positive early endosomes accumulate in neurons early during disease progression, even before there is deposition of amyloid beta [65], and a similar build-up of early endosomes is seen in SN DA neurons of patients with iPD [9]. Given that LRRK2 colocalizes with EEA1-positive early endosomes [66], it is possible that the observed increase in LRRK2 kinase activity in iPD is responsible for the accrual of Rab5- and transferrin-positive endosomes iPD [9]. Consistent with this hypothesis, blocking LRRK2 activity with PF-360 prevented the accumulation of Rab5-positive early endosomes in the rotenone model of iPD [9]. While Rab5b, and not Rab5a has been nominated as a potential substrate for LRRK2 in overexpression studies [25,67,68], available antibodies cannot distinguish between different Rab5 isoforms or their phosphorylation state. Nevertheless, if Rab5b is a LRRK2 substrate, this could explain the observed accumulation of Rab5 in iPD DA neurons [9]. However, Rab5b has yet to be validated as a bona fide LRRK2 substrate in an endogenous context. It is possible that prolonged LRRK2 kinase activity leads to general disruption of early to late endosomal maturation by preventing recruitment of the Rab5-specific GDI. Supporting this interpretation, the number of late endosomes and lysosomes was normalized by LRRK2 kinase inhibition in a rodent model of iPD [9]. Further investigation is necessary to determine if early increases in LRRK2 activity coincide temporally with accumulation of Rab5 (endosomes) and whether this might serve as an early indicator for iPD pathology.

The balance between lysosomal repair and clearance can be dysregulated in PD. This imbalance is functionally significant because extensive damage to the outer lysosomal membrane is detrimental to cell health. To circumvent harmful consequences that might arise from membrane permeabilization, damaged lysosomes can be tagged for autophagic removal in a process called lysophagy, which is initiated by recognition of cytosolic lectins, such as Galectin-3, at the lysosome membrane. However, endosomal sorting complexes required transport (ESCRT) machinery can also be recruited to damaged or permeabilized lysosomes before lysophagy machinery, in an attempt to seal and repair damaged lysosomal membranes [69,70]. New evidence suggests that LRRK2 can recruit Rab8A and the ESCRT-III component, charged multivesicular body protein 4B (CHMP4B), to damaged lysosomes, regulating the decision between membrane repair and lysophagy [48]. Any disruption in ESCRT recruitment can interrupt lysosome repair and result in irreversible lysosome damage, which can be lethal to the cell [69]. Interestingly, macrophages from carriers of LRRK2-PD mutations (G2019S or R1441C) accumulate vesicles that are positive for Rab8A and Galectin-3, suggesting that increased LRRK2 activity disrupts the balance between lysosomal membrane repair and lysophagy [48].

It is apparent that pathological LRRK2 kinase activity can disrupt membrane and vesicle trafficking and might affect lysosomal function deleteriously (Figure 2). Therefore, a deeper understanding of the factors that can activate aberrant LRRK2 kinase activity, and the mechanisms thereof, is essential.

Box 1. The endolysosomal pathway

The endolysosomal pathway uses a series of cargo vesicles to internalize nutrients and recycle or degrade cargo (via lysosomes; Figure I). Newly internalized cargo from the plasma membrane is rapidly targeted to early endosomes and shuttled to either the distinct thin tubular or the large luminal microdomain of the early endosome. The tubular microdomain of the early endosome is involved in recycling cargo either back to the plasma membrane via recycling endosomes or shuttling cargo to the TGN via retrograde transport. These two distinct pathways sometimes involve the retromer complex. Cargo in the large luminal microdomain of the early endosome eventually matures into the late endosome (also known as multivesicular bodies). This maturation process is marked by Rab5 to Rab7 conversion, in which Rab5-enriched early endosomes become Rab7-enriched (acidic) late endosomes. As cargo is trafficked toward lysosomes, the luminal pH of endosomes gradually decreases. The endosomal sorting complexes required transport (ESCRT) machinery facilitates late endosomal trafficking and fusion of these mature acidic late endosomes with lysosomes, in which proteolytic degradation of cargo occurs. Given that recycling or degrading endosomes are regulated by distinct Rab GTPases, and a subset of these Rab GTPases are substrates of LRRK2, LRRK2 is believed to regulate aspects of the endolysosomal pathway.

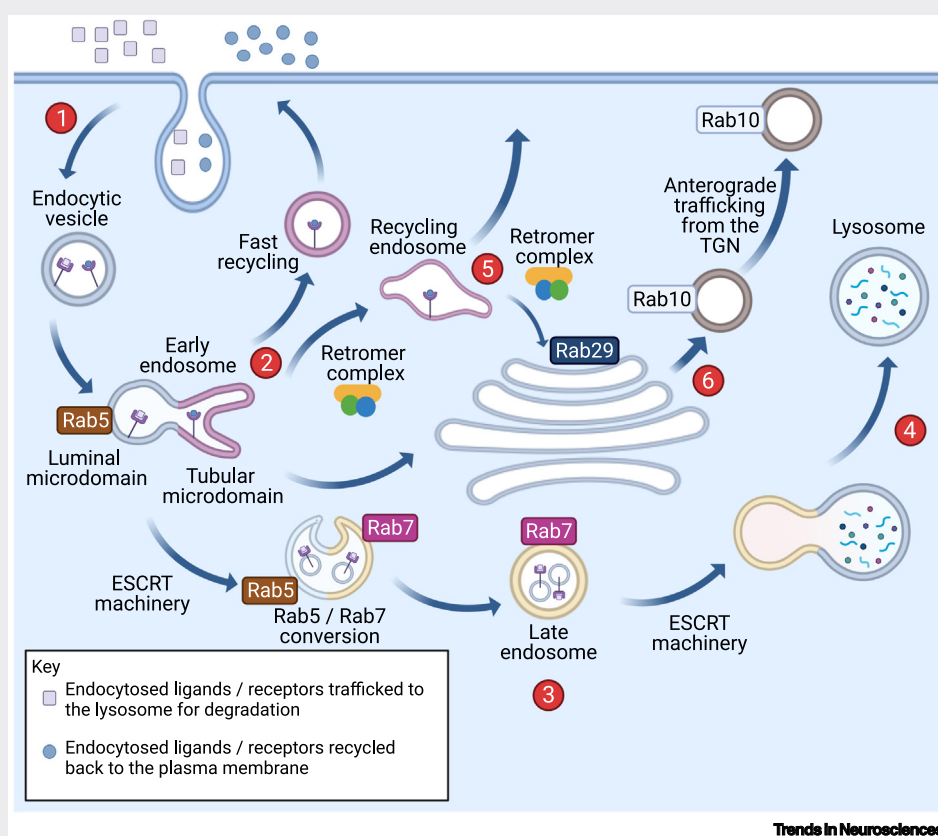
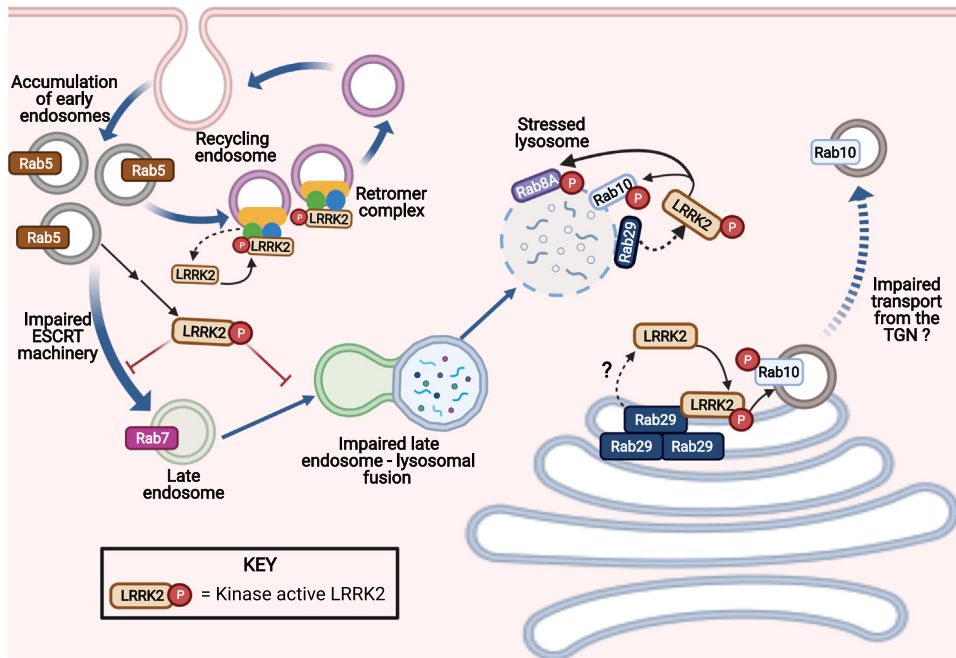


Figure I. Schematic of endolysosomal pathways. (1) From plasma membrane endocytosis to the formation of early endosomes. This is subsequently followed by two divergent pathways: (2) formation of recycling endosomes and (3) maturation to late endosome and eventual (4) lysosomes. (5) The retromer complex can traffic cargo to the *trans*-Golgi network (TGN) and (6) supports anterograde vesicular trafficking from the TGN. Abbreviation: ESCRT, endosomal sorting complexes required transport.

LRRK2: activation by oxidative and endolysosomal stress

Dopaminergic neurons are highly vulnerable to oxidant stress. This enhanced sensitivity to **oxidative stress** might be due, in part, to DA itself, which can auto-oxidize and lead to the accumulation of free radicals and the highly reactive DA **quinone**. Accumulation of these toxic free radicals and quinones can lead to mitochondrial dysfunction, reactive oxygen species



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Figure 2. The role of leucine-rich repeat kinase-2 (LRRK2) in endolysosomal trafficking. In Parkinson's disease (PD), elevated LRRK2 activity can lead to accumulation of Rab5-positive early endosomes as well as impairment in the ESCRT machinery. These deficits can, in turn, prevent early to late endosomal maturation; as a result, there are fewer Rab7-positive late endosomes. These deficits in vesicle trafficking can also cause an increase in LRRK2 activity, albeit through an undefined mechanism(s). Impaired late endosomal-lysosomal fusion caused by aberrant LRRK2 kinase activity can result in stressed lysosomes, which then recruit and activate LRRK2, possibly via Rab29. At a stressed lysosome, active LRRK2 can recruit and phosphorylate Rab8A and Rab10. The biological significance of this recruitment has yet to be fully characterized but likely involves lysosomal vesicle formation/budding and exocytosis. At the recycling endosome, LRRK2 can be activated by the retromer complex. Although the mechanism responsible for this recruitment is unknown, it is clear that such recruitment can result in improper retrograde trafficking from the recycling endosome to the *trans*-Golgi network (TGN). Moreover, at the TGN, high levels of Rab29 (in overexpression systems) can recruit and activate LRRK2, which results in accumulation of Rab10-positive vesicles at the TGN, which in turn, can lead to impaired anterograde trafficking. Abbreviations: ESCRT, endosomal sorting complexes required transport.

(ROS) production, and oxidatively damaged proteins [71]. Therefore, nigrostriatal DA neurons rely heavily on surveillance mechanisms (e.g., **autophagy**) and the endolysosomal pathway to identify and remove dysfunctional proteins, organelles, and unwanted material to maintain homeostasis. Disruption of these surveillance pathways can occur preferentially in DA neurons, because DA and its oxidation products can cause lysosomal and mitochondrial dysfunction [64,71–73].

LRRK2 is a **reduction/oxidation** (redox)-sensitive protein and elevated oxidative stress can exert its effects, in part, through enhanced LRRK2 kinase activity (Figure 3, Key figure). Several reports indicate that LRRK2 kinase activity is increased by H_2O_2 [7,74,75] or decreased by the antioxidant curcumin [74]. The kinase activity of wild-type endogenous LRRK2 can be elevated dose dependently by physiological concentrations of H_2O_2 , and this can be blocked by the ROS scavenger α -tocopherol [7]. Others have reported that H_2O_2 increases wild-type LRRK2 kinase activity to the same extent as the G2019S mutation [75]. Moreover, there is strong evidence that LRRK2 can be modulated by: (i) the major ROS producer, NADPH oxidase 2; (ii) exposure to the toxicant rotenone; and (iii) overexpression α -synuclein, all of which promote the formation of ROS [7]. Thus, redox state might be an important regulator of LRRK2 kinase activity. In this regard, the kinase domain shows a high degree of homology with other redox-

Key figure

Activation of wild-type leucine-rich repeat kinase-2 (LRRK2) kinase

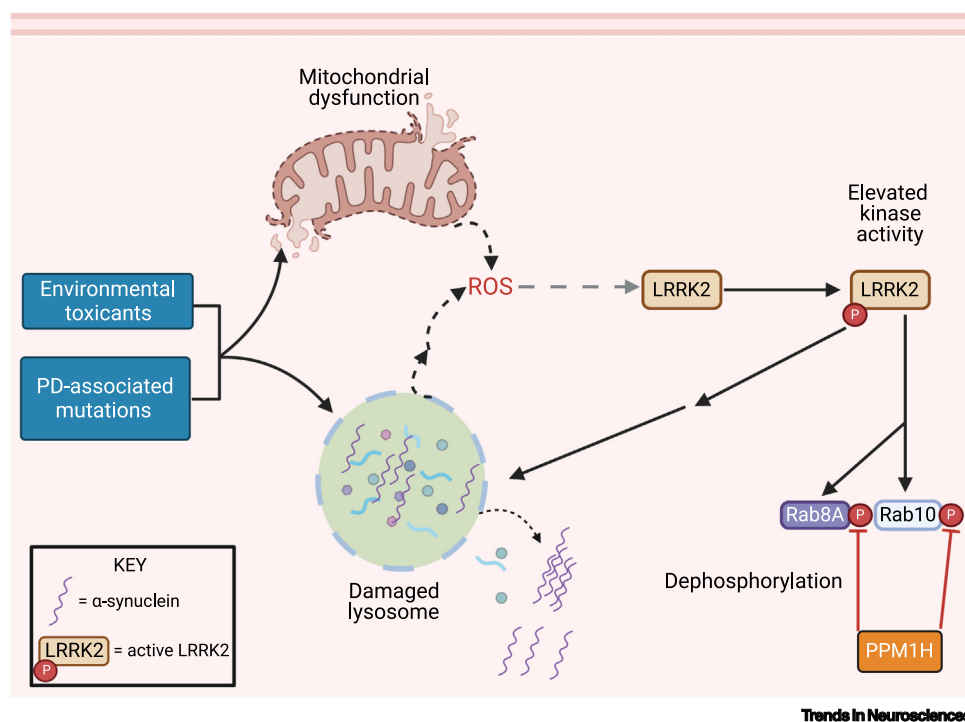


Figure 3. Environmental contaminants linked to Parkinson's disease (PD) [82] and PD-associated risk genes [2] affect mitochondrial function and the endolysosomal system, either directly or indirectly. Interestingly, mitochondrial dysfunction and endolysosomal stress are key mediators of LRRK2 kinase activation. A convergent event of both mitochondrial and lysosomal dysfunction is the generation and accumulation of reactive oxygen species (ROS), which, in turn, can activate LRRK2. Whether the ROS build-up from damaged lysosomes is released directly from these lysosomes or is a downstream consequence is undefined. Enhanced LRRK2 kinase activity, as a result of either mitochondrial dysfunction or stressed lysosomes, can further damage lysosomes and result in membrane permeabilization and leakage of damaged cargo and proteins, including α -synuclein into the cytoplasm. This then results in the toxic accumulation and aggregation of α -synuclein. The phosphatase PPM1H, which is known to counteract LRRK2 activity by dephosphorylating specific Rab proteins (e.g., Rab8A and Rab10) might normalize aberrant LRRK2 kinase activity. However, this signaling axis remains to be explored.

sensitive MAP-kinases [76]. Cysteine residues on proteins can readily undergo oxidative modifications, which, in turn, can influence function and enzymatic activity. The kinase activation loop of LRRK2 contains two solvent-exposed vicinal cysteine residues. These residues are conserved across many species [77] and are not present in LRRK1, suggesting that they have a specific role in LRRK2 activation. It is plausible that these cysteine residues are important for sensing the redox state of the local subcellular environment and function to activate LRRK2 [18]. If so, it might be possible to design LRRK2 inhibitors that target them [78]. Going forward, exploration of the redox sensitivity of LRRK2 could yield novel insights into the molecular mechanisms that drive its kinase activity.

While LRRK2 is regulated by oxidative stress, it might also enhance oxidative stress [7,79], and it is possible that increased LRRK2 kinase activity might indirectly damage components of the

endolysosomal pathway. In addition to oxidative stress, endolysosomal and autophagic stressors also activate LRRK2. For example, pleiotropic trafficking deficits induced by **monensin**, and lysosomal membrane rupture caused by **chloroquine** or **L-leucyl-L-leucine-methyl ester** can induce LRRK2 activation in several different cell types [46,48,53,54]. Although whether these apparently disparate stimuli share similar molecular mechanisms of LRRK2 activation is unknown, interfering with endolysosomal or autophagic function can lead to secondary oxidative stress [80–82]. Conversely, oxidative stress and ROS can cause impairment of endolysosomal function [81], which highlights the reciprocal interplay between redox signaling and autophagy.

Conceivably, LRRK2-mediated lysosomal stress might cause mitochondrial deficits, perhaps via impaired mitophagy, thereby leading to increased mitochondria-derived ROS, but the sequence and directionality of this signaling remain to be defined. Interestingly, blocking LRRK2 kinase activity *in vivo* using PF-360 prevented substantia nigra pars compacta (SNc) DA neuron endolysosomal deficits caused by rotenone, an oxidative stressor [9]. Thus, endolysosomal impairment can activate LRRK2, and LRRK2 kinase activity contributes to endolysosomal dysfunction.

LRRK2: a target of environmental toxicants

The mechanisms underlying aberrant LRRK2 activity and its downstream effects might be influenced by exogenous factors (Figure 3). In this context, an obvious, but often overlooked, factor involved in LRRK2 kinase activation is environmental contaminants linked to PD [83], which are increasingly reported to enhance LRRK2 kinase activity in the brain [7,84–87]. Epidemiological data show increased risk for iPD among pesticide-exposed farmers and residents of agricultural or rural communities [88,89], as well as industrial workers exposed to chlorinated solvents [90,91]. The pesticides paraquat and rotenone are both associated with a 2.5-fold increased risk for PD [92]. Similarly, twin studies show that organic solvent trichloroethylene (TCE) increases PD risk (odds ratio 6.1) [91]. Thus far, the mechanisms that enhance the risk for developing iPD following toxicant exposure appear to involve accumulation of ROS and/or mitochondrial dysfunction.

Emerging evidence indicates that LRRK2 might link mitochondrial toxicant-driven oxidative stress with PD-related pathologies. For example, when rats were chronically exposed to the mitochondrial complex I inhibitor, rotenone, or the industrial solvent, TCE, there was oxidative damage in the SN, increased LRRK2 kinase activity, phosphorylated Rab10 accumulation, and endolysosomal dysfunction, with resultant selective nigrostriatal DA neuron degeneration [9,84]. Co-treatment with the selective LRRK2 kinase inhibitor, PF-360, prevented all rotenone-induced pathologies, including DA neuron degeneration in rats [9].

Peripheral inflammation might be important in PD pathogenesis and LRRK2 is highly expressed in immune cells [93]; thus, exposure to exogenous toxicants that influence the kinase activity of LRRK2 might also contribute to excessive neuroinflammation in PD. Support for this idea comes from a study using macrophages treated with manganese, and which also suggests non-neuronal cells as an important target of toxicant-induced LRRK2 kinase activity [94,95].

LRRK2: therapeutic strategies

Therapeutic approaches for LRRK2 activation stemming from environmental toxicant exposure

The molecular mechanisms by which LRRK2 is activated by environmental contaminants remain to be delineated. Clearer understanding of these mechanisms could lead to new therapeutic and prophylactic measures. Inhibition of LRRK2 by small-molecule kinase inhibitors or **antisense oligonucleotides** (ASOs) might protect against LRRK2-mediated pathology induced by

environmental contaminants, as seen experimentally in the rotenone and pre-formed fibril α -synuclein rodent models of PD [9,10]. Thus, individuals who have had a known exposure to an environmental toxicant linked to increased PD risk and display elevated LRRK2 kinase activity in a biological sample (e.g., blood neutrophils, PBMCs, or urinary exosomes) might represent a population in which to attempt to prevent the onset, or slow the progression, of PD. Thus, LRRK2 inhibitors might be a unique therapeutic avenue for individuals exposed to PD-linked environmental toxicants. However, at a minimum, this approach would require integration of environmental monitoring, validation of LRRK2 biomarkers, and off-label usage of approved LRRK2 therapeutics. Accordingly, the bar to mitigate neuropathology caused by environmental toxicant-induced LRRK2 kinase activity is high and will require further characterization and validation in human subjects.

Therapeutic approaches targeting LRRK2 for iPD

In contrast to the toxicant exposure mitigation discussed previously, the use of LRRK2 therapeutics for iPD disease modification will likely require long-term, possibly life-long, treatment after initiation of therapy. Given the expression of LRRK2 in tissues other than the brain, especially the lungs, there is justified concern about the potential for ‘on-target’ (LRRK2-specific) toxicity, such as pulmonary toxicity. There are several lines of evidence that, at least in our view, argue against the possibility of severe on-target toxicity. A recent study in nonhuman primates examined the pulmonary effects of three structurally distinct LRRK2 kinase inhibitors and found only mild histopathological lung changes that were reversible and without noticeable effect on lung function [96]. Studies in mice and rats also yielded reassuring results [38,97–99]. Importantly, two recent genetic studies demonstrated that LRRK2 loss-of-function variants in humans do not appear to have negative effects on health or survival [100,101], which further argues in favor of the potential safety of LRRK2-targeted therapeutic approaches.

At the time of writing, a Phase Ib clinical trial to evaluate the safety, target engagement, and effects on putative biomarkers of a LRRK2 kinase inhibitor (BIIB122/DNL151) in subjects with PD with or without LRRK2 mutations had recently completed.¹ Additionally, an antisense oligonucleotide therapeutic targeting LRRK2 is entering Phase I clinical studies (NCT03976349)ⁱⁱ and is open to subjects with LRRK2 mutations as well as those with iPD. However, a consideration to keep in mind is that only a fraction of LRRK2 is believed to be active at a given time and it remains uncertain that reducing overall LRRK2 levels will reduce the amount of LRRK2 kinase activity. Going forward, as more is learned about the structure, function, and biology of LRRK2, it is likely that additional approaches to target LRRK2 therapeutically will emerge.

Concluding remarks

Despite current uncertainties and knowledge gaps (see [Outstanding questions](#)), it is becoming increasingly clear that LRRK2 has a central role in the pathogenesis of iPD. Common variants in the LRRK2 locus are known risk factors for iPD. Oxidative stress and endolysosomal impairment, both of which have critical roles in PD, activate LRRK2 kinase activity. Similarly, environmental toxicants, implicated epidemiologically in PD risk, activate LRRK2. Moreover, *in vivo* blockade of LRRK2 activity is protective in genetic and toxicant models of PD. As noted, early-stage clinical trials of LRRK2-based therapeutics are already underway. Given that PD is the fastest growing neurological disease in the world [102] and that LRRK2 is central to PD pathogenesis, a deeper understanding of the cellular biology of LRRK2 and its activation mechanisms is essential.

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Outstanding questions

What are the biological functions of LRRK2, and how do they differ by cell type?

The phosphatase PPM1H counters the effects of LRRK2. Does dysregulated PPM1H contribute to LRRK2-driven pathologies?

What are the molecular mechanisms by which oxidative stress and endolysosomal/autophagic stress activate wild-type LRRK2, and are they the same?

Does dopamine or its oxidation products activate LRRK2 in SNc neurons?

What is the most efficacious and safe way to mitigate aberrant LRRK2 kinase activity in PD: small-molecule kinase inhibitors, RNAi, or other approaches?

Inhibition of LRRK2 kinase activity appears to prevent neurodegeneration *in vivo*, but can it slow or stop degeneration after the pathogenic cascade of PD has been initiated?

Given that PD-associated environmental toxicants (e.g., rotenone, paraquat, and TCE) activate LRRK2 kinase activity, can LRRK2 therapeutics be used to mitigate postexposure risk of developing PD?

Are individuals with LRRK2 loss-of-function mutations less likely to develop PD, and are such individuals relatively resistant to PD-associated environmental toxicants?

Could assessment of peripheral LRRK2 kinase activity provide the basis for a reliable biomarker for disease state, patient stratification, or target engagement?

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Declaration of interests

None declared by authors.

Resources

ⁱ<https://denaltherapeutics.gcs-web.com/node/8141/pdf>

ⁱⁱ<https://clinicaltrials.gov/ct2/show/NCT03976349>

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